

Optimising Daylight for Human Wellbeing: in Appenzell Innerhoden & Ausserhoden

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Abstract

Daylight plays a central role in human health and wellbeing. Beyond allowing us to see, it regulates our circadian rhythm, mood, alertness, and cognitive function. In mountainous regions like Appenzell (Innerhoden and Ausserhoden), traditional tall houses and complex topographies present unique daylight challenges. This project proposes a novel Fuzzy-AI daylight simulation framework to evaluate and optimise daylight penetration into residential spaces under variable seasonal and weather conditions. The study aims to provide window design guidelines that maximise daylight exposure, preserve architectural heritage, and promote occupant wellbeing.

1 Introduction

Daylight is more than illumination—it is a biological and psychological necessity. Exposure to natural light regulates the human circadian rhythm, enhances mood, and improves sleep quality (Boubekri et al. 2014). In Appenzell, houses are typically tall, with vertically oriented facades and steep roofs adapted to the Alpine climate. However, this verticality often limits daylight access to lower levels, especially during winter months when sunlight angles are low.

This research integrates daylight simulation, fuzzy logic, and architectural design analysis to identify optimal window sizes and placements that improve daylight distribution in tall Appenzell houses. The findings aim to guide architects and urban planners in designing human-centred, daylit spaces that balance energy efficiency and wellbeing.

2 Daylight and Human Wellbeing

Numerous studies show that sufficient daylight exposure enhances both physical and psychological health. Morning light exposure synchronises the circadian system, helping regulate hormone release, metabolism, and sleep patterns (Rathod et al. 2023). Insufficient daylight or prolonged exposure to artificial light has been associated with seasonal affective disorder (SAD), fatigue, and reduced cognitive performance (Editorial 2021).

Well-designed windows are therefore vital not only for visual comfort but also for maintaining a healthy biological rhythm. The objective of this project is to ensure that daylight distribution in indoor environments

aligns with human wellbeing needs while respecting regional architectural traditions.

3 Appenzell Architectural Context

The architecture of Appenzell reflects both climatic adaptation and cultural identity. Traditional houses are characterised by:

- Steep, multi-storey facades with limited window openings.
- Wooden structures and deep eaves that provide shade but restrict light.
- Orientation often constrained by mountainous terrain, affecting daylight access.

While these designs offer insulation and protection, they also reduce interior daylight levels—especially on lower floors. Adapting traditional forms to optimise daylight while preserving aesthetic integrity requires precise, evidence-based design strategies.

4 Fuzzy-AI Daylight Simulation with Uncertainty

Current daylight simulation tools such as Radiance or ClimateStudio generate deterministic outputs (e.g., “200 lux at 10 a.m.”). However, daylight availability in mountainous regions is inherently uncertain—affected by cloud cover, season, and topographic shadows.

This project introduces a fuzzy-AI framework that integrates uncertainty directly into daylight predictions. Instead of crisp results, the model outputs probabilistic statements such as:

“The probability of achieving sufficient daylight between 8–10 a.m. is high (0.8).”

This approach combines:

1. **Fuzzy logic:** to handle uncertain daylight inputs (e.g., diffuse light levels).
2. **AI-based simulation:** using adaptive learning to adjust for real weather data.
3. **Architectural geometry analysis:** evaluating how window height, width, and orientation influence light penetration.

The simulation framework allows architects to virtually test different window configurations and instantly visualise their daylight probability scores across seasons.

5 Window Design Recommendations

Based on preliminary simulations and existing daylight literature, several design strategies are proposed for improving daylight in Appenzell houses:

- **Increase window height rather than width:** Tall, narrow windows allow light to penetrate deeper into rooms and distribute illumination more evenly.
- **Integrate clerestory openings:** High-level windows capture sunlight even during low sun angles in winter.

- **Orient key living spaces toward the southeast:** Maximises morning light exposure, supporting circadian alignment.
- **Use high-transmittance glazing:** Improves daylight quality while reducing reliance on artificial lighting.
- **Preserve traditional aesthetics:** Daylight optimisation should be achieved through proportional adjustments and light-reflecting materials rather than drastic facade changes.

These guidelines can be applied in both new constructions and retrofits, supporting sustainable and health-conscious building design.

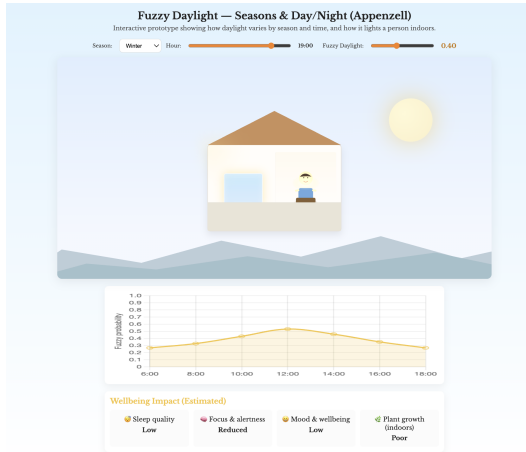
6 Benefits for Human Wellbeing

Enhancing daylight access contributes directly to human wellbeing. Adequate natural light improves mood, reduces eye strain, and strengthens circadian rhythm regulation (Barbosa et al. 2022). In community spaces such as schools or offices, well-lit environments have been linked to higher productivity and satisfaction (Smith et al. 2020).

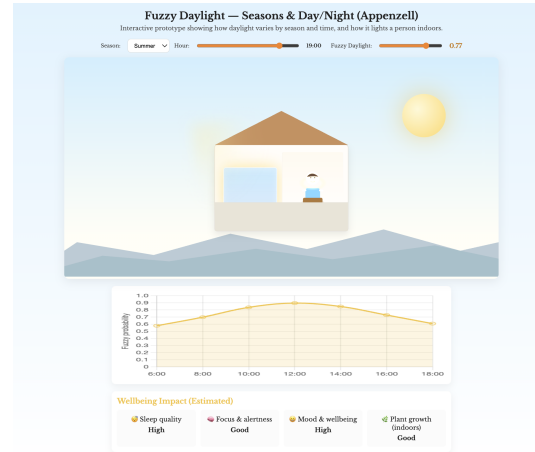
For the residents of Appenzell, optimising daylight through smart window design can mitigate winter light deprivation, enhance comfort, and promote a deeper connection between architectural form and natural environment.

7 Visual Illustration: How Daylight Impacts Wellbeing

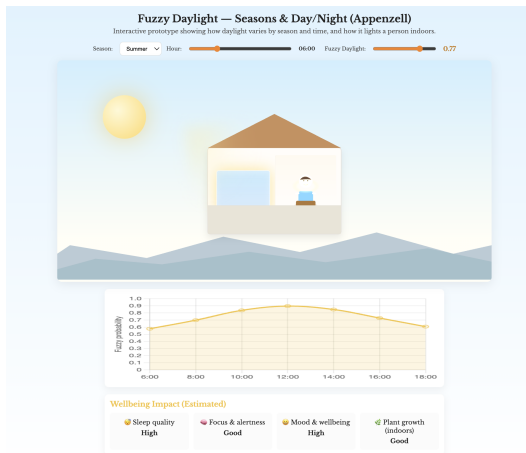
To complement the theoretical framework, the following numbered illustrations (1–8) demonstrate how daylight influences human wellbeing — physiologically, psychologically, and socially. The figures also serve as a conceptual guide for designing window placements in traditional Appenzell homes.



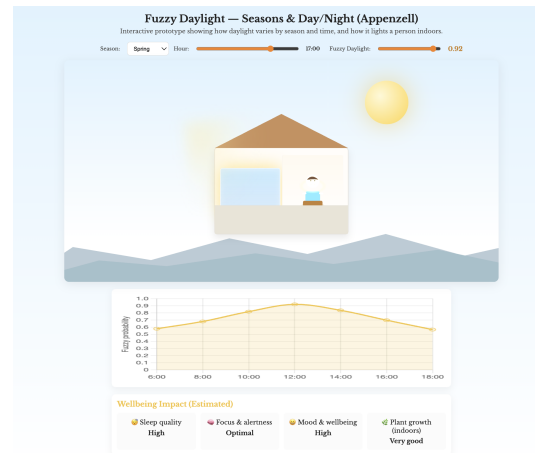
(1) How Daylight Impacts Wellbeing.



(2) How Daylight Impacts Wellbeing.

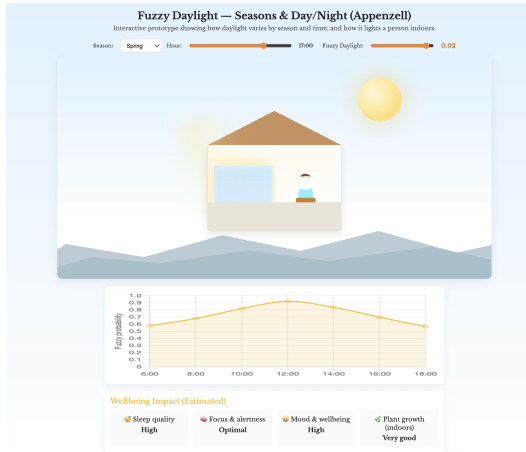


(3) How Daylight Impacts Wellbeing.

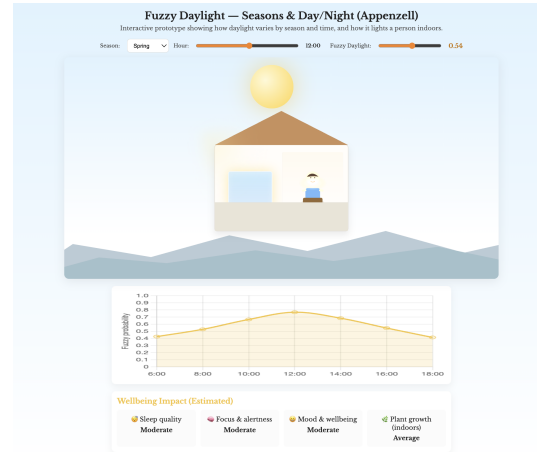


(4) How Daylight Impacts Wellbeing.

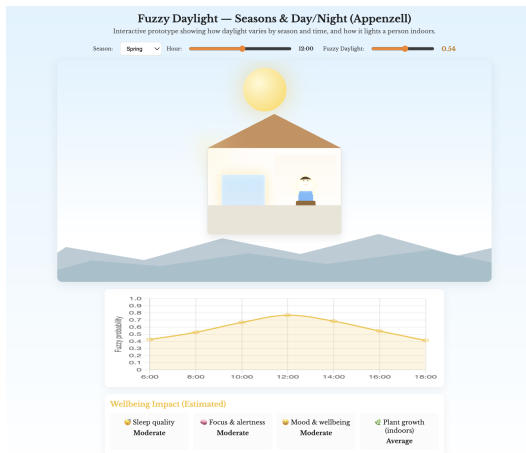
Figure 1: Illustrations (1–4) showing the psychological and social benefits of daylight exposure in built environments.



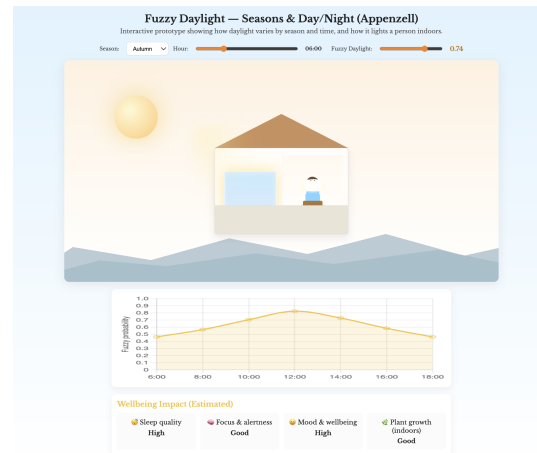
(5)How Daylight Impacts Wellbeing.



(6)How Daylight Impacts Wellbeing.



(7)How Daylight Impacts Wellbeing.



(8)How Daylight Impacts Wellbeing.

Figure 2: Illustrations (5–8) showing physiological and environmental effects of daylight, especially in the context of Appenzell’s traditional elevated architecture.

Design Considerations for Maximum Daylight: The architectural geometry of a building plays a decisive role in daylight access. In regions such as Appenzell, where homes are often positioned on sloped terrain with varying sun angles, the alignment and proportions of windows significantly influence indoor illuminance. Figures 9–11 present essential design principles for optimising daylight — from proper window orientation and height placement to the use of reflective surfaces that amplify natural brightness without glare.

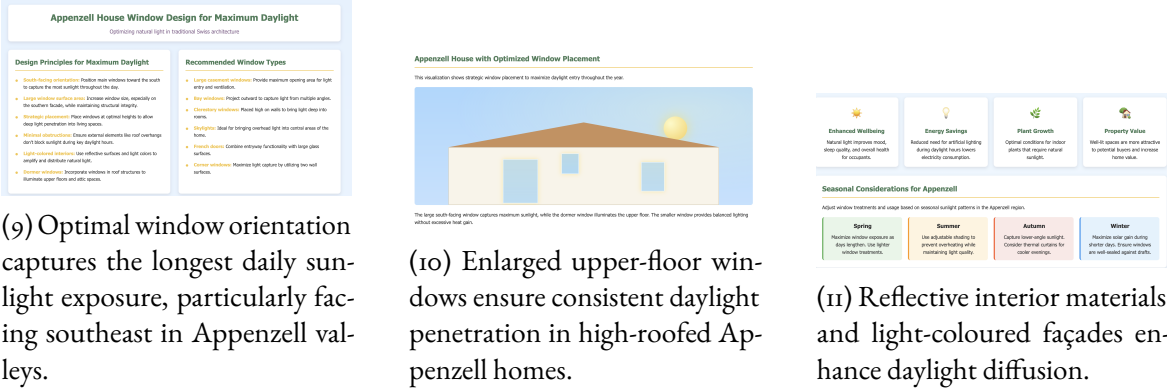


Figure 3: Illustrations (9–11) showing architectural and environmental strategies for maximising daylight in traditional Appenzell houses. These include orientation, window design, and reflective surface choices that promote human wellbeing and energy efficiency.

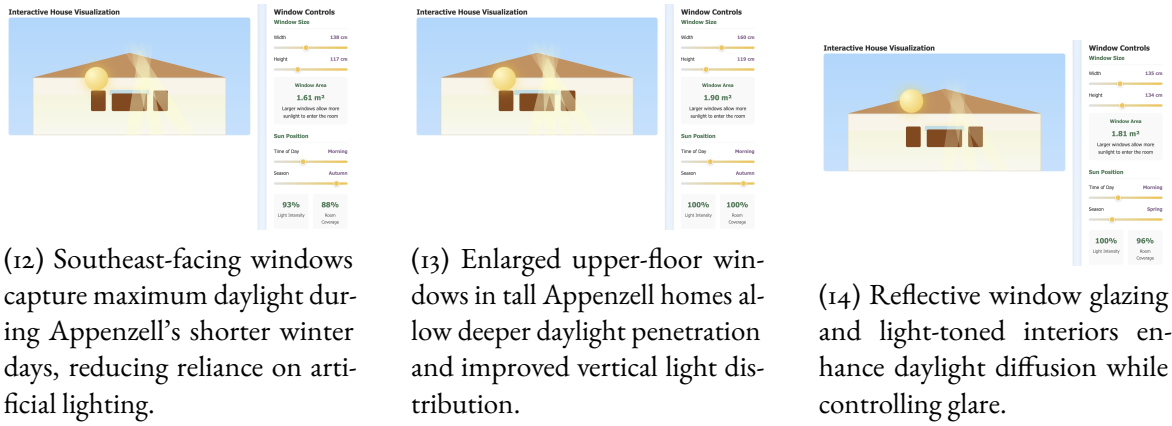
Windows sizes according to house area:

In the Appenzell cantons, traditional houses are typically multi-storey, constructed with steep roofs and elevated foundations to adapt to the region’s hilly terrain. These architectural forms influence how daylight enters interior spaces across different floor levels and seasons. To ensure optimal natural lighting, window dimensions must be adapted not only to the aesthetic character of the houses but also to their total floor area and surrounding landscape.

Empirical daylight studies suggest that the ideal **window-to-floor ratio (WFR)** ranges between **20% and 35%** in mountain regions, depending on orientation and shading. For instance:

- Smaller homes (under 100 m²) benefit from larger window openings — approximately **30–35% WFR** — to counter limited internal reflection surfaces.
- Medium houses (100–180 m²) typically achieve balance at **25–30% WFR**.
- Large homes (above 180 m²) perform efficiently at around **20–25% WFR**, especially when supplemented by roof windows or clerestories.

Appenzell’s unique geography — with east–west facing valleys and frequent morning mist — encourages slightly higher south-facing window ratios to maximise morning and midday light. By contrast, north-facing façades are generally designed with smaller apertures to reduce heat loss while maintaining visual comfort.



(12) Southeast-facing windows capture maximum daylight during Appenzell's shorter winter days, reducing reliance on artificial lighting.

(13) Enlarged upper-floor windows in tall Appenzell homes allow deeper daylight penetration and improved vertical light distribution.

(14) Reflective window glazing and light-toned interiors enhance daylight diffusion while controlling glare.

Figure 4: Figures (12–14) highlight window design principles that optimise daylight in Appenzell homes. Proper window orientation, size, and reflective surface treatment improve light quality, support visual comfort, and promote wellbeing across seasons.

8 Conclusion

This project integrates fuzzy logic, daylight simulation, and architectural research to develop adaptive design recommendations for improving daylight access in Appenzell's traditional houses. By quantifying uncertainty and connecting architectural form to wellbeing outcomes, the project aligns with the Velux Stiftung's mission to explore daylight's multifaceted role in human life.

The expected outcome is a scientifically grounded yet culturally sensitive daylight design framework that enhances wellbeing, respects heritage, and fosters sustainable living.

References

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